Övning 16 - Plasticitet och utmattning

Effektivspänning

- Mått på "styrkan" av spänningsfältet.

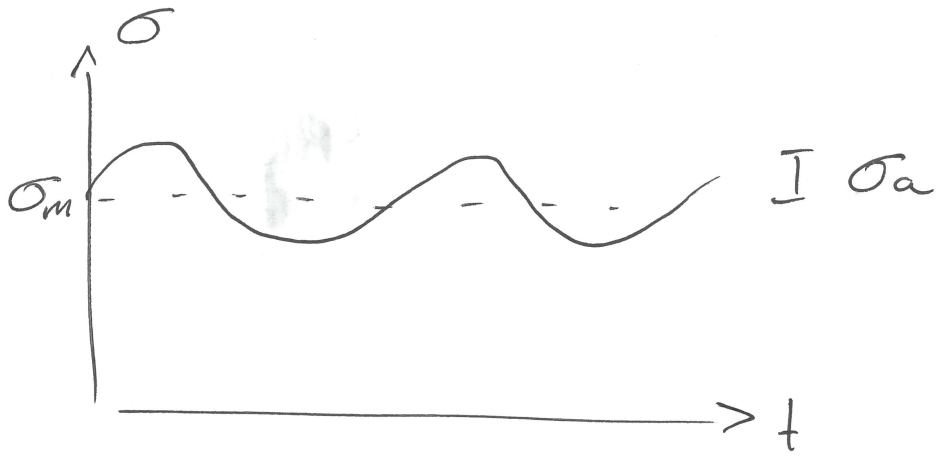
Von-Mises

$$\sigma_e^{vM} = \left(\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - \sigma_x\sigma_y - \sigma_x\sigma_z - \sigma_{yz} + 3\tau_{xy}^2 + 3\tau_{xz}^2 + 3\tau_{yz}\right)^{\frac{1}{2}}$$

Tresca

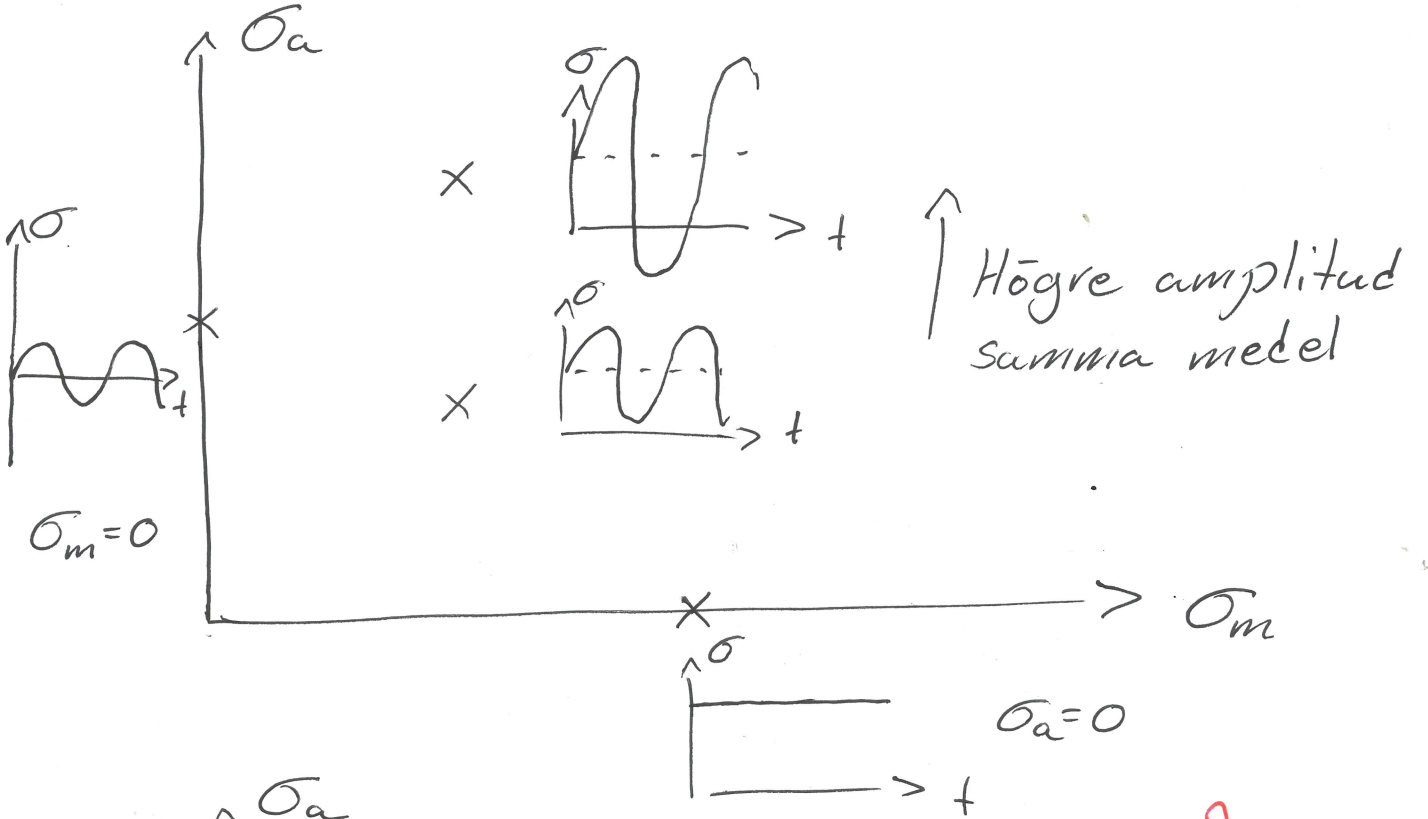
$$\sigma_e^{Tr} = \max\left(|\sigma_1 - \sigma_2|, |\sigma_1 - \sigma_3|, |\sigma_2 - \sigma_3|\right)$$

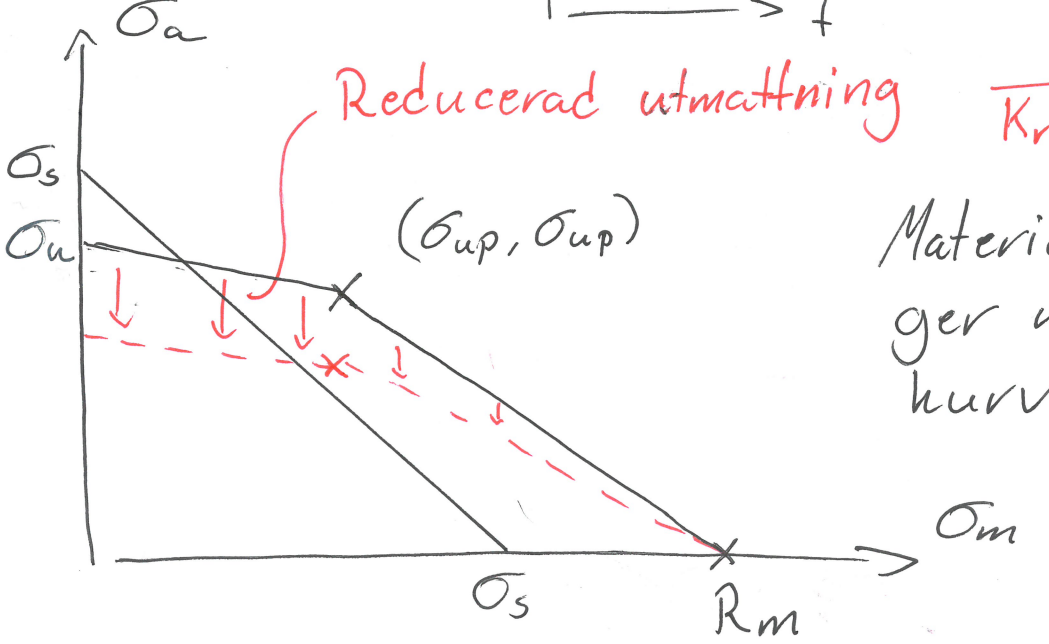
Plasticering vid $\sigma_e \geq \sigma_s$



σ_m - Medelspänning

σ_a - Amplitudspänning





2.11.1

$\sigma_x = 70\,MPa \quad \sigma_y = 120\,MPa \quad \tau_{xy} = 60\,MPa$

Sökt: Effektivspänning a) Tresca

b) von Mises

a) Ta fram huvudspänningar

$$\det(\underset{\sim}{S} - \sigma \underset{\sim}{I}) = \det \begin{bmatrix} 70-\sigma & 60 & 0 \\ 60 & 120-\sigma & 0 \\ 0 & 0 & -\sigma \end{bmatrix} = 0$$

$$\Rightarrow (70-\sigma)(120-\sigma)(-\sigma) + 60 \cdot 60\sigma = 0$$

$$\Rightarrow \sigma^3 - 190\sigma^2 + 4800\sigma = 0$$

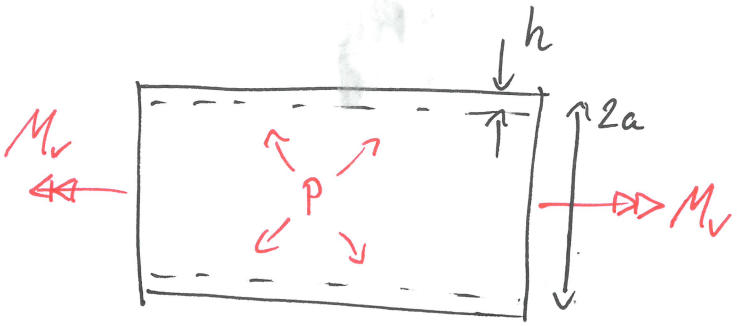
$$\sigma_A = 0$$

$$\Rightarrow \sigma^2 - 190\sigma + 4800 = 0$$

$$\Rightarrow \sigma_B = 30 \qquad \sigma_C = 160$$

$$\Rightarrow \begin{cases} \sigma_1 = 160\,MPa \\ \sigma_2 = 30\,MPa \\ \sigma_3 = 0\,MPa \end{cases}$$

2.11.7



Sträckgräns σ_s

vid enbart inre tryck

$\sigma_e^{vM} = \frac{\sigma_s}{2}$

Sökt: Vridande moment för plasticering

Beräkna von-Mises vid enbart inre tryck.

$$\sigma_r = 0 \qquad \sigma_\varphi = \frac{Pa}{h} \qquad \sigma_z = \frac{Pa}{2h}$$

$$\sigma_e^{vM} = \left(\frac{P^2a^2}{h^2} + \frac{P^2a^2}{4h^2} - \frac{P^2a^2}{2h^2} \right)^{\frac{1}{2}} = \frac{\sqrt{3}\,Pa}{2h} = \sigma_s \frac{1}{2}$$

$$\Rightarrow \boxed{P = \frac{\sigma_s h}{\sqrt{3}\,a}}$$

Totalt $\sigma_r = 0$ $\sigma_\varphi = \frac{Pa}{h}$ $\sigma_z = \frac{Pa}{2h}$ $\tau_{\varphi z} = \frac{M_v}{2\pi a^2 h}$

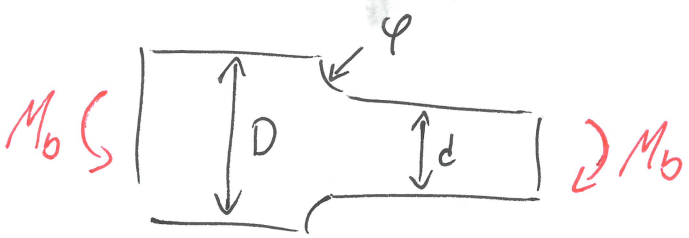
$$\sigma_e^{vM} = \left(\frac{P^2a^2}{h^2} + \frac{P^2a^2}{4h^2} - \frac{P^2a^2}{2h^2} + 3\frac{M_v^2}{4\pi^2 a^4 h^2} \right)^{\frac{1}{2}}$$

$$= \left(\frac{\sigma_s^2}{4} + \frac{3M_v^2}{4\pi^2 a^4 h^2} \right)^{\frac{1}{2}} = \sigma_s$$

$$\Rightarrow \sigma_s^2 = \frac{\sigma_s^2}{4} + \frac{3M_v^2}{4\pi^2 a^4 h^2} \Rightarrow \boxed{M_v = \sigma_s \pi a^2 h}$$

2.12.32

SS1650-01



$D = 80\,mm$

$d = 60\,mm$

$\varphi = 5\,mm$

Sökt: Säkerhet mot utmattning n_u

Profildjup $R_t = 100\,\mu m$

$M_b(t) = M_0(1 + 2\sin(\omega t))$

$M_0 = 1000\,Nm$

Haigh-diagram

① Ideal kurva $R_{p0.2}$, R_m, σ_{ub}, σ_{ubp}

FS s. 386
14/1650-1

$$\Rightarrow \begin{cases} R_{p0.2} = 310\,MPa \\ R_m = \frac{590+690}{2} = 640\,MPa \\ \sigma_{ub} = \pm 270\,MPa \\ \sigma_{ubp} = 240 \pm 240\,MPa \end{cases}$$

② Reducerad utmattnings kurva (s. 295)

Bestäm $\frac{\lambda}{K_r K_f K_d}$ λ, K_r, K_f K_d

Ej gjuten produkt $\lambda = 1$

K_r från fig 25.12

$$\begin{cases} R_a = \frac{R_t}{4} = 25\,\mu m & [FS\ 25.10] \\ R_m = 640\,MPa \Rightarrow K_r = \frac{1}{0.7} \end{cases}$$

2.12.32 del 2

Anvisning finns $K_d = 1$ $K_f = 1 + q(K_t - 1)$

q - Känslighetsfaktor [FS:s. 298]

Fig 25.9 $\left\{ \begin{array}{l} \rho = 5mm \\ R_m = 640 MPa \end{array} \right\} \Rightarrow q = 0.85$

K_t - Formfaktor [Fs: s 367]

tab 32:4.1 $\left\{ \begin{array}{l} \rho/d = 5/60 = 0.083 \\ D/d = 80/60 = 1.33 \end{array} \right\} \Rightarrow K_t = 1.7$

$\Rightarrow K_f = 1 + 0.85(1.7 - 1) \Rightarrow \boxed{K_f = 1.6}$

Reduktionsfaktor (Bara för σ_a)

$$\frac{\lambda}{K_r K_f K_d} = \frac{1}{\frac{1}{0.7}\ 1.6\ 1} = 0.4375$$

$$\Rightarrow \left\{ \begin{array}{l} R_m = 640 MPa \\ \sigma_{ub,red} = \pm 118 MPa \\ \sigma_{ubp,red} = 240 \pm 105 MPa \end{array} \right.$$

2.12.32 del 3

③ Arbetslinje

Böjning $\sigma(t) = \frac{M(t)}{I} z_{max} = \frac{32M(t)}{\pi d^4}$

($z_{max} = \frac{d}{2}$, $I = \frac{\pi d^4}{64}$)

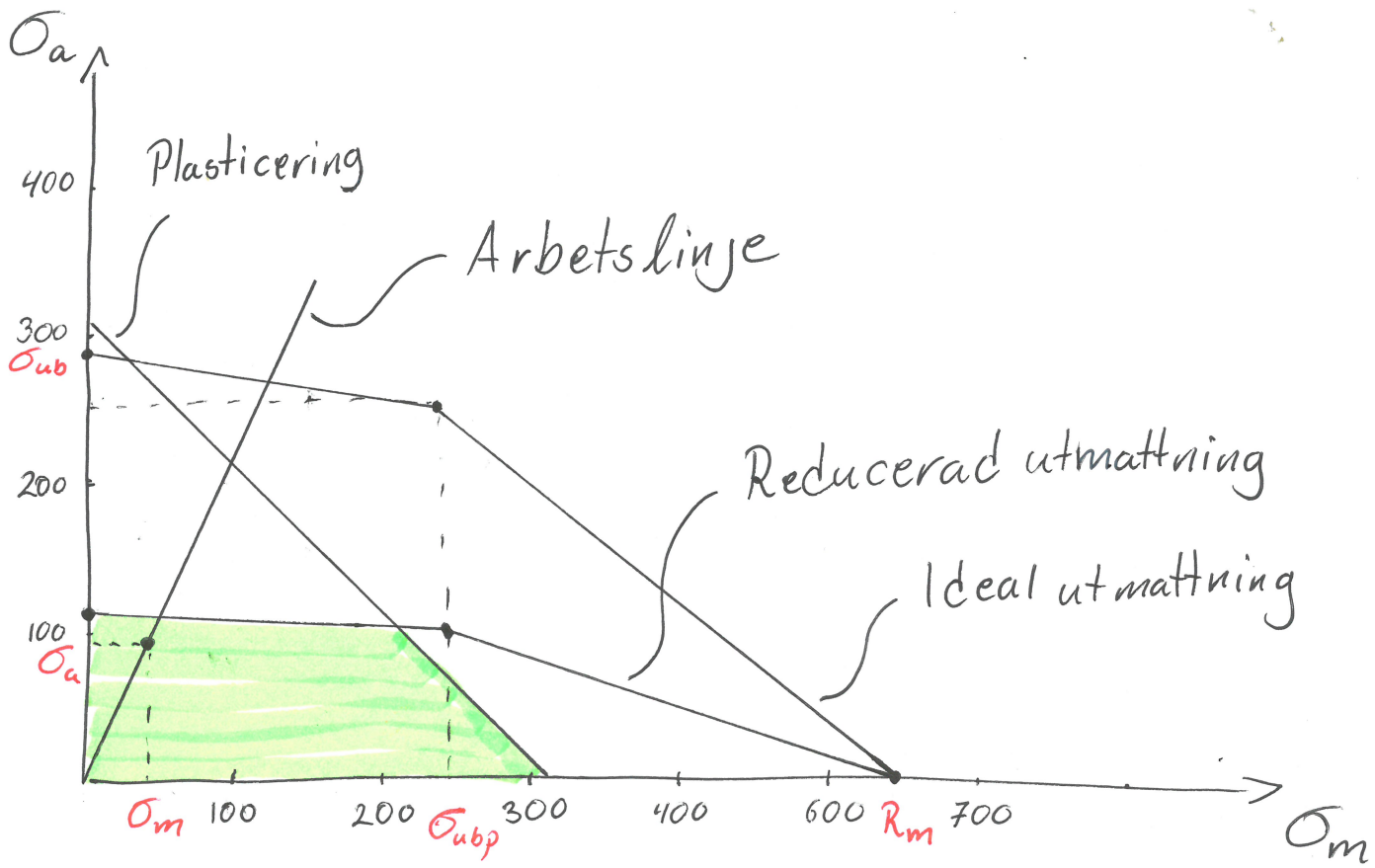
$$\sigma(t) = \frac{32 M_0 (1 + 2\sin\omega t)}{\pi d^4}$$

$$\sigma_m = \frac{32 M_0}{\pi d^4} = 47\,MPa$$

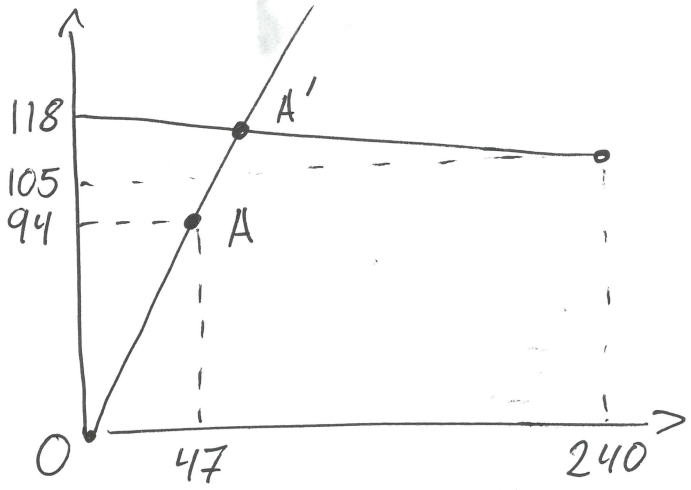
$$\sigma_a = \frac{64 M_0}{\pi d^4} = 94\,MPa$$

Arbetslinjen ges som $\sigma_a(\sigma_m)$

I det här fallet $\boxed{\sigma_a = 2\sigma_m}$



2.12.32 del 4



$$n_{ut} = \frac{|OA'|}{|OA|}$$ säkerhetsfaktor

A' är skärningspunkten mellan två linjer

linje 1 $\sigma_a = 118 + \frac{105-118}{240}\sigma_m$.

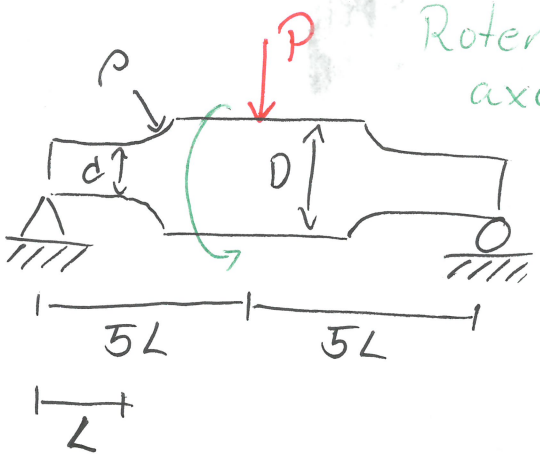
linje 2 $\sigma_a = 2\sigma_m$

$$\Longrightarrow \begin{cases} \sigma_a' = 57.44 \text{ MPa} \\ \sigma_m' = 114.89 \text{ MPa} \end{cases}$$

$$n_{ut} = \frac{\sqrt{57.44^2 + 114.89^2}}{\sqrt{94^2 + 47^2}} = 1.22$$

$$n_{ut} = 1.22$$

2.12.41



Roterande axel

SS1550-01

$D = 140\,mm$

$d = 100\,mm$

$\rho = 10\,mm$

$L = 20\,mm$

Profildjup $R_t = 10\,\mu m$

Sökt: Säkerhet mot utmattning n_{ut}

Ideal kurva

FS s. 386

141550-01

$$\Rightarrow \begin{cases} R_{p0.2} = 270\,MPa \\ R_m = \frac{490+590}{2} = 540\,MPa \\ \sigma_{ub} = \pm 240\,MPa \\ \sigma_{ubp} = 210 \pm 210\,MPa \end{cases}$$

Reducerad uttmattningskurva (s. 295)

Två olika punkter måste behandlas

(I) Vid anvisning

(II) Under lasten

λ - Ej gjutet $\lambda_I = \lambda_{II} = 1$

2.12.41 del 2

Anvisning → (I) Ja

Anvisning → (II) Nej

(I) $\boxed{K_{dI} = 1}$ $K_{f,I} = 1 + q(K_t - 1)$

q fig 25.9 $\Rightarrow \begin{cases} \rho = 10\,mm \\ R_m = 540\,MPa \end{cases} \Rightarrow q = 0.86$

K_t tab 32.4.1 $\Rightarrow \begin{cases} \rho/d = 0.1 \\ D/d = 1.4 \end{cases} \Rightarrow K_t = 1.65$

$\Rightarrow \boxed{K_{fI} = 1.559}$

(II) $K_{fII} = 1$

K_{dII} fig 25.10 $\Rightarrow \begin{cases} D = 140\,mm \\ R_m = 540\,MPa \end{cases} \Rightarrow K_{dII} = 1.14$

Ytfinhet fig 25.12

$K_{rI} = K_{rII} \Rightarrow \begin{cases} R_a = \frac{R_t}{4} = 2.5\,\mu m \\ R_m = 540\,MPa \end{cases} \Rightarrow K_{rI} = K_{rII} = 1.05$

(I) $\dfrac{\lambda_I}{K_{dI} K_{fI} K_{rI}} = 0.609$ (II) $\dfrac{\lambda_{II}}{K_{dII} K_{fII} K_{rII}} = 0.836$

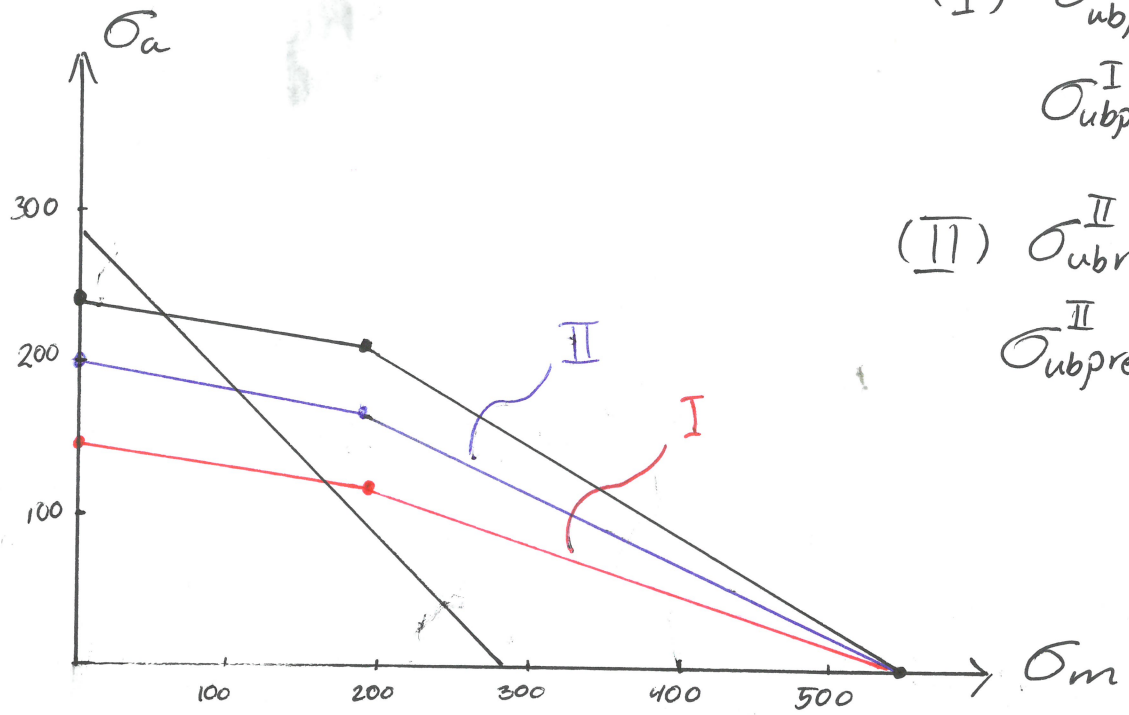
2.12.41 del 3

(I) $\sigma^{I}_{ub,red} = \pm 146\,MPa$

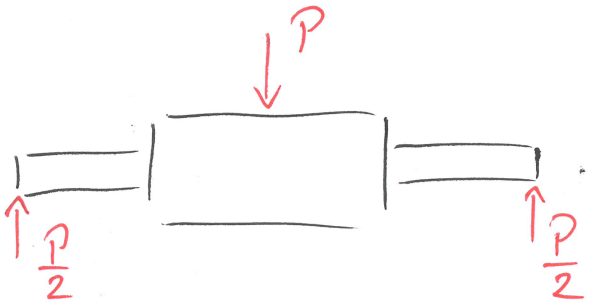
$\sigma^{I}_{ubpred} = 210 \pm 128\,MPa$

(II) $\sigma^{II}_{ubred} = \pm 201\,MPa$

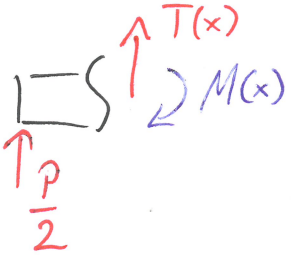
$\sigma^{II}_{ubpred} = 210 \pm 176\,MPa$



Spänningar för arbetslinjen -
Behöver moment för punkterna vid anvisning och under kraften



$0 < x < 5L$



$\Rightarrow M(x) = -\frac{Px}{2}$

vid anvisning $x = L$ $M_I = -\frac{PL}{2}$

Under lasten $x = 5L$ $M_{II} = -\frac{5PL}{2}$

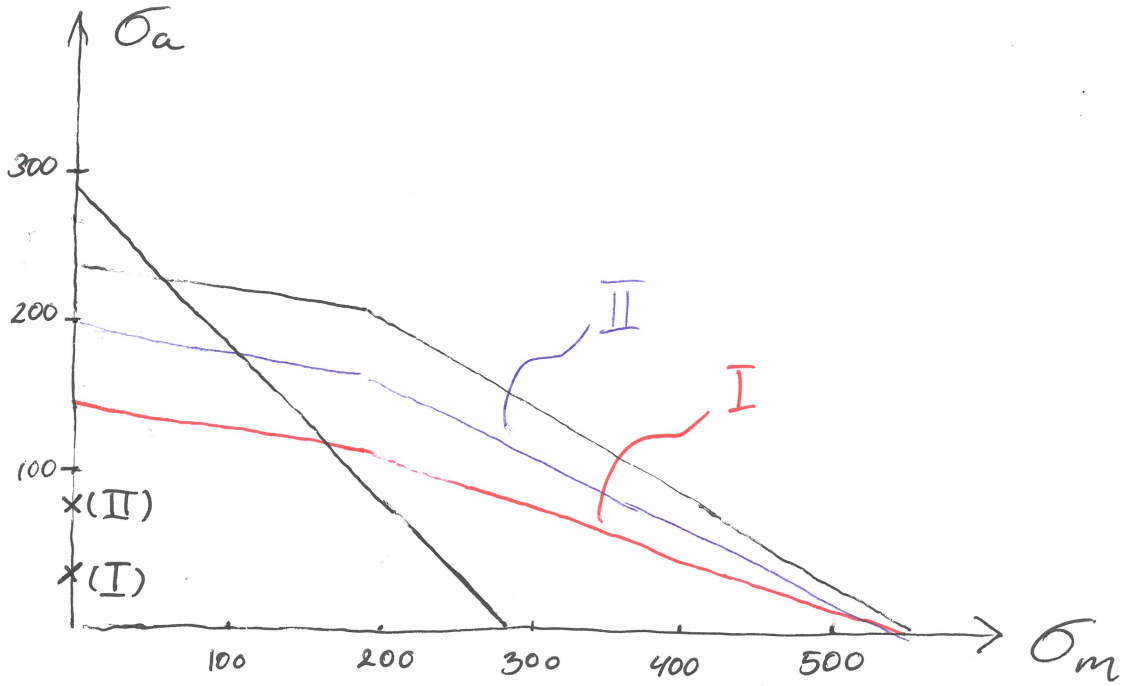
2.12.41 del 4

Spänningar $\sigma_x = \frac{M(x)}{I} z$ $\quad I = \frac{\pi d^4}{64}$ $\quad z = \frac{d}{2}$

Vid anvisning $\sigma_I = \frac{-\frac{PL}{2}}{\frac{\pi d^4}{64}} \frac{d}{2} = 40.7\,MPa$

Under lasten $\sigma_{II} = \frac{-\frac{5PL}{2}}{\frac{\pi D^4}{64}} \frac{D}{2} = 74.2\,MPa$

Eftersom axeln roterar är lasten rent växlande ($\sigma_m = 0$)



$n_I = \frac{146}{40.7} = 3.58$

$n_{II} = \frac{201}{74.2} = 2.7$